CSC1042 Computer Network II

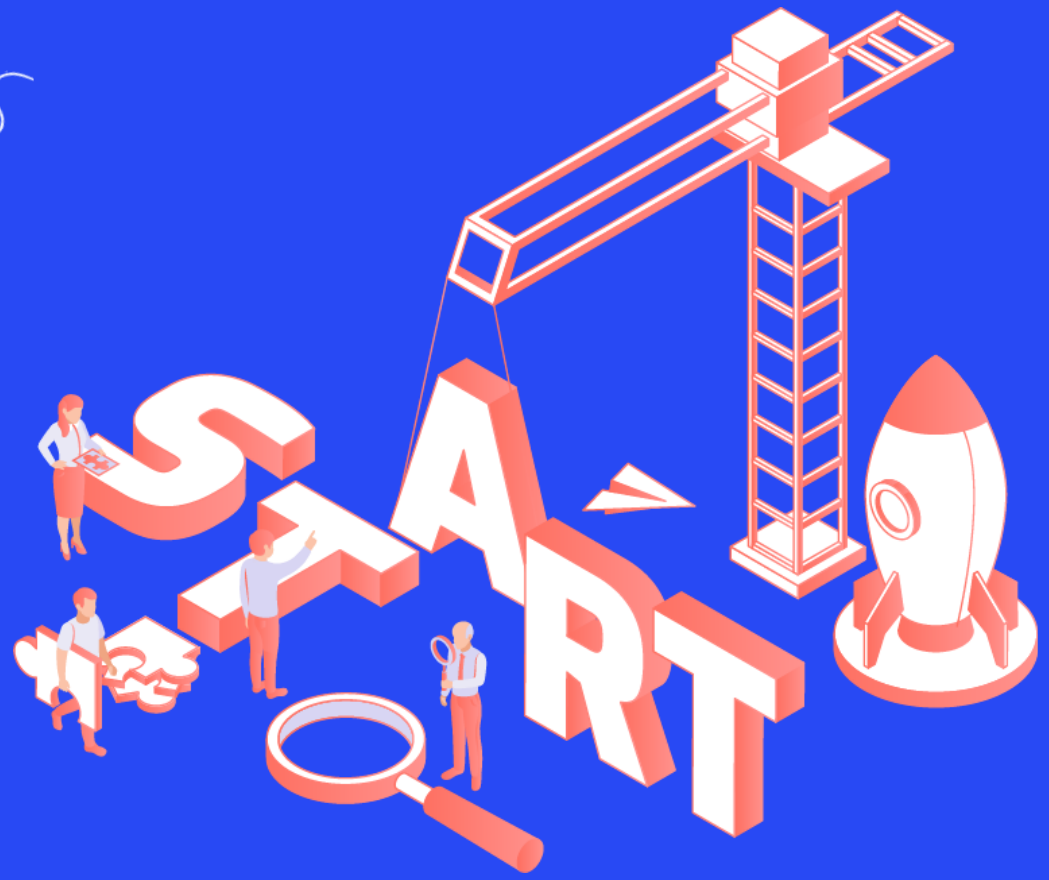
IPv6

Dr. Mohammed Amine TOGOU



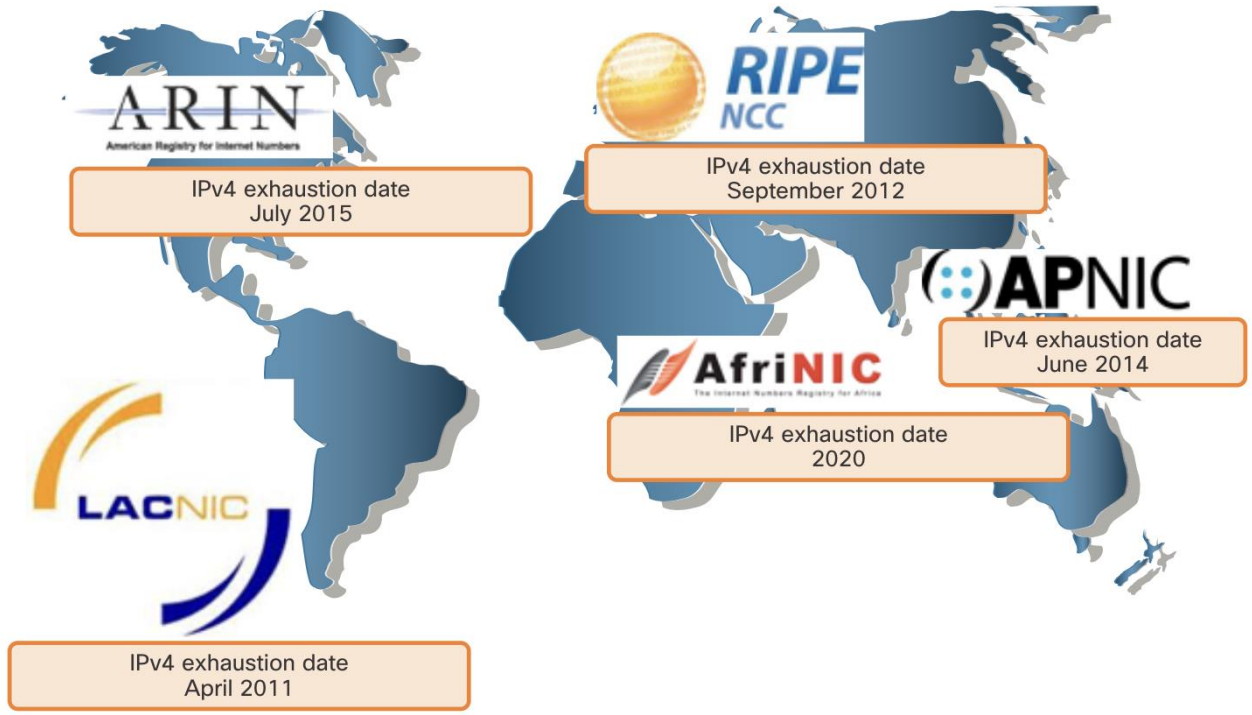
Learning outcomes

- Explain the need for IPv6
- Explain how IPv6 addresses are represented
- Compare types of IPv6 addresses
- Describe IPv6 header



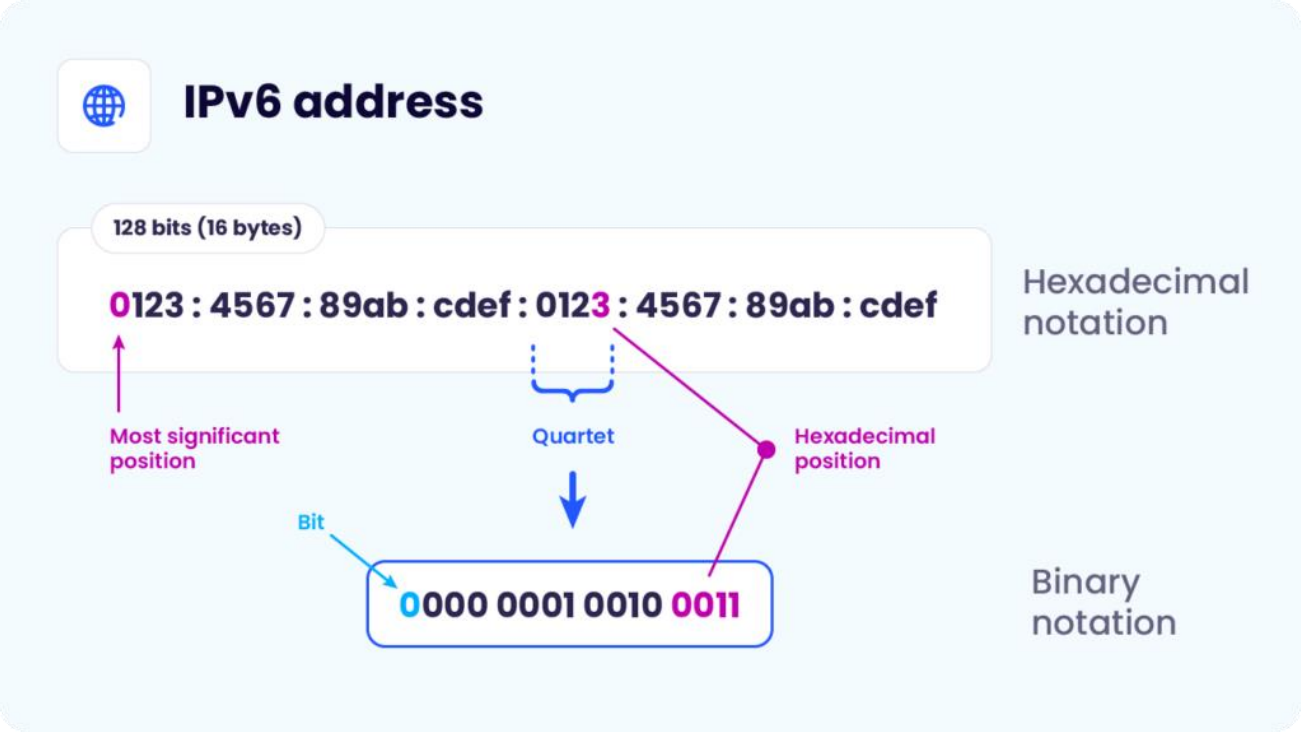
Need for IPv6

- IPv4 run out of addresses to be assigned
- IPv6 was proposed to solve the issue of depleted addresses
- IPv6 provides a new way of addressing on the internet
- It also adds some features to mitigate the limitation of IPv4



IPv6 addresses

- Each IPv6 address is 128-bit long (16 bytes)
- They are written in 8 blocks of 2-bytes, called quartets
- Each block is made of 4 hex digits
- Blocks are separated by :



IPv6 addresses

- IPv6 can be abbreviated using the following rules:
  - Leading 0s in any quartet can be removed
  - Blocks of 0s can be replaced with ::
    - Can be done only once

0000:0000:0000:0000:0000:0000:0000:0001



0:0:0:0:0:0:0:1



::1

Exercise

- Abbreviate the following IPv6 address

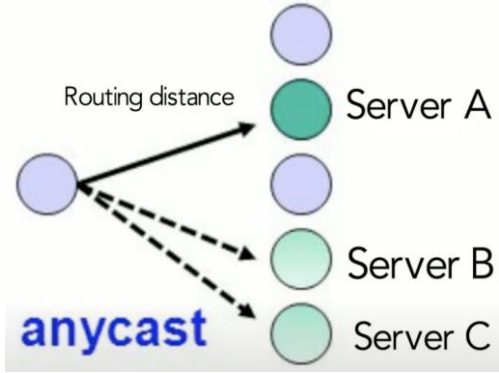
0fd3:0000:5e4a:b290:0000:0000:700a:00df

Types of IPv6 addresses

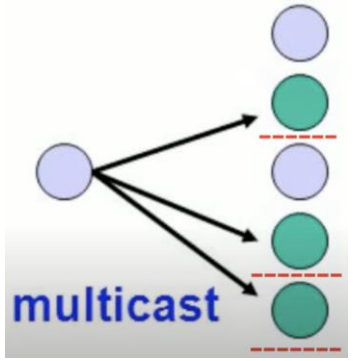
No broadcast!
Replaced with multicast

Unicast: represents a single interface. Packets are delivered to a single host

Anycast: identifies one or more interfaces, e.g., servers that support the same function can use the same IP address. Packets sent to that IP address are forwarded to the nearest server. Used for load-balancing



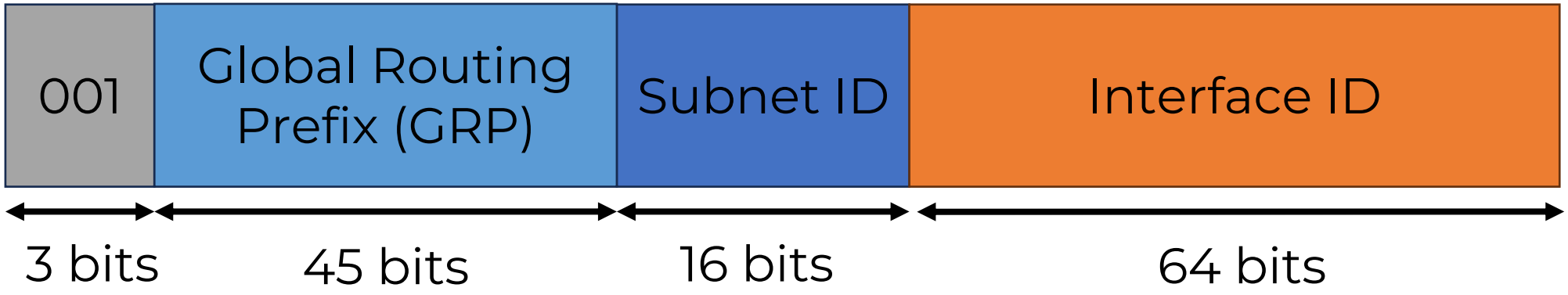
Multicast: represents a dynamic group of hosts. Packets sent to this address are delivered to many hosts



Unicast addresses

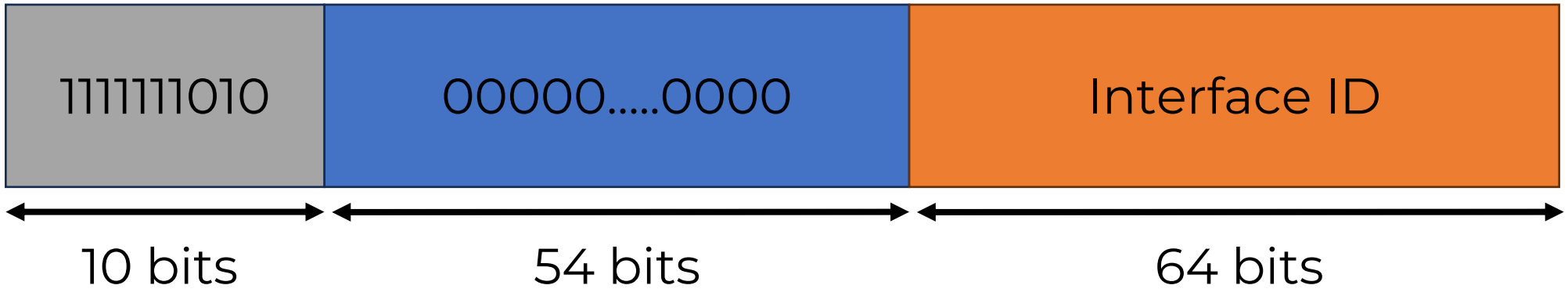
- Global
  - addressable on the Internet (like IPv4 public addresses)
  - They start with 2000::/3
- Link Local
  - Required for every IPv6-enabled device
  - Used for sending packets over the local subnet. Packets from these addresses are not forwarded by routers to other subnets.
  - They start with fe80::/10
- Unique Local
  - Used for sending packets between devices within the site
  - They start with fd00::/8
  - Not routable on the Internet

Global unicast address structure



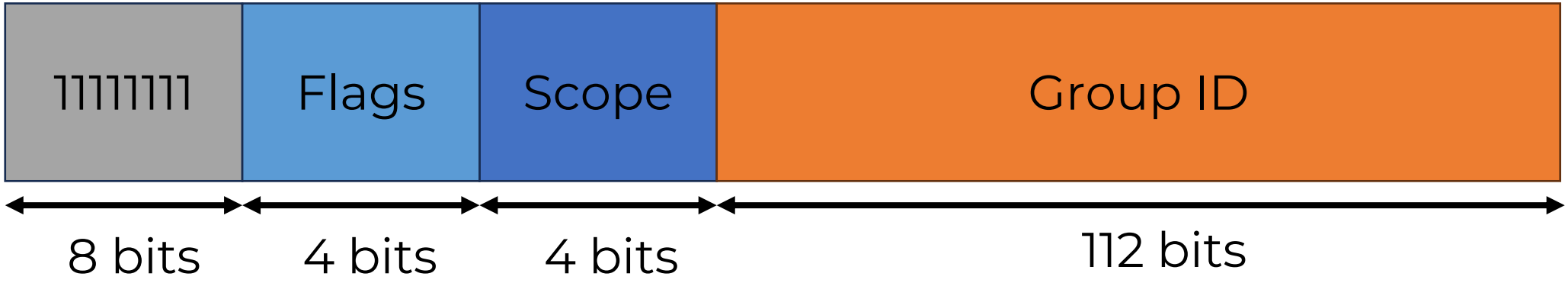
- The first 48 bits are for the ISP.
  - First 3 bits always set to 001 (2000::/3 in hex)
  - GRP is assigned by the Internet Authority
- 16 bits for the Subnet ID.
  - This allows network admin to create a hierarchy of subnets within the organisation and assign unique values to them
- 64 bits for the interface ID, i.e., the host

Link-local address structure



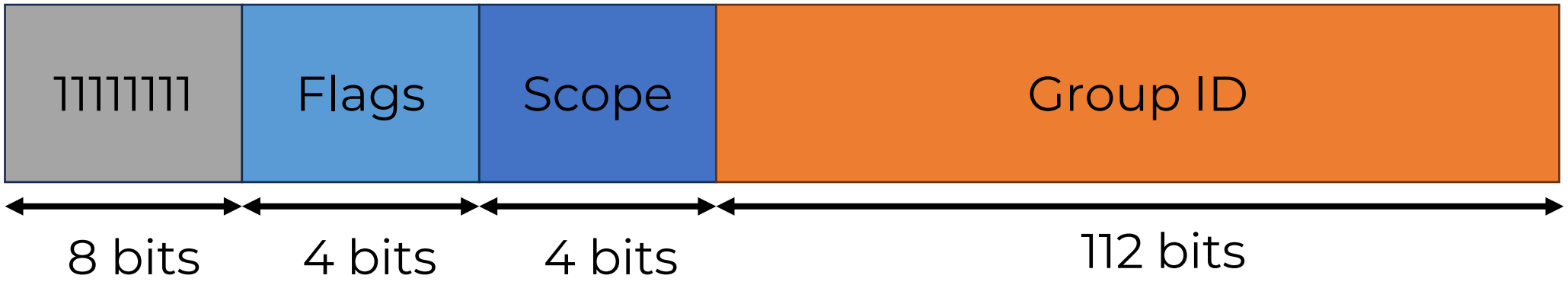
- First 10 bits always set to 1111111010 → fe80::/10
- A Link-Local address must be assigned to every network interface on which the IPv6 protocol is enabled
- A host can automatically derive its own Link-Local IP address, or the address can be assigned manually

Multicast address structure



- First 8 bits always set to 11111111 → ff00::/8
- 4 Flags: 0 (reserved), R (rendezvous point), P (network prefix), T (transient)
  - T = 0 → permanent multicast address assigned by the Internet authority
  - T = 1 → dynamic multicast address assigned by servers (e.g., MADCAP) or applications themselves.

Multicast address structure



- Scope indicates the IPv6 network for which the multicast traffic is intended
- Routers use the Scope field to determine if multicast traffic can be forwarded
- e.g., addresses that begin with ff02::/16 are intended to stay on the local link

Value (hex)	Scope
1	Node
2	Link
5	Site
8	Organisation
E	global

Unspecified and Loopback addresses

- Unspecified address: 0:0:0:0:0:0:0:0
  - It indicates the absence of an address
  - Used by a host as source address field of an IPv6 packet before it is assigned an address
- Loopback address: 0:0:0:0:0:0:0:1
  - Must not be assigned to any physical network interface
  - Equivalent to 127.0.0.1 in IPv4

IPv4 and IPv6 co-existence

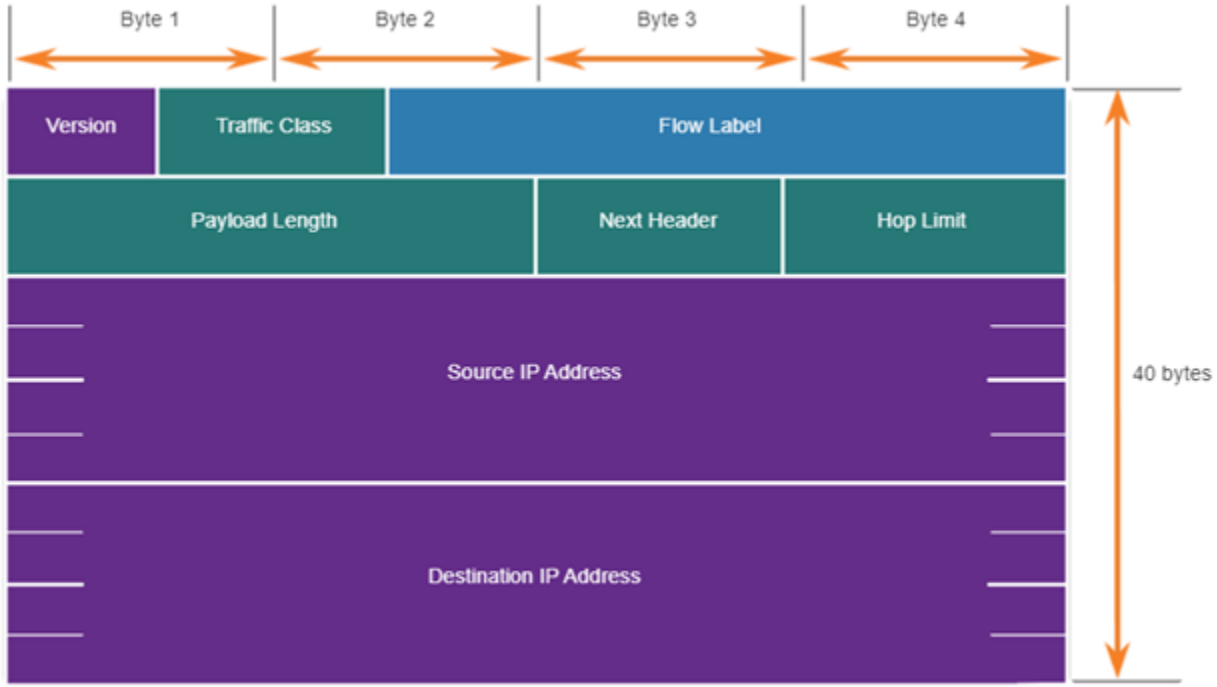
- IPv4 addresses can be translated into IPv6 addresses
- Write the IPv4 address in the last 32 bits of the IPv6 address

80 bits	16 bits	32 bits
00000….0000	ffff	IPv4 address (in hex)

e.g., 136.206.48.94 → ::ffff:88ce:305e

IPv6 header

- Version: indicate the version of IP (v6)
- Traffic Class: equivalent to type of service in IPv4 header
- Flow Label: used so that all packets of the same flow receive the same type of handling by routers
- Payload Length: indicates the size of the data portion
- Next Header: equivalent to the Protocol IPv4 field
- Hop Limit: equivalent to TTL in IPv4 header
- Source and Destination IPv6 addresses



IPv6 enhancements over IPv4

- IPv6 makes finding addresses on the network easier
- Neighbour Discovery Protocol (NDP) replaces ARP along with ICMP redirect and router discovery services
- Hosts use NDP to discover routers as well as auto-configure their addresses
- It is also used for Duplicate Address Detection (DAD)

inform a host of a more optimal route through a network

enables hosts to locate routers on the local subnet and use them as gateways

Gratuitous ARP does the same for IPv4

IPv6 enhancements over IPv4

- NDP uses ICMPv6 messages, originated from nodes on link-local with hop limit of 255
  - This is a security feature
  - Routers will discard any Neighbour Discovery message with a TTL < 255
  - This prevents “fake” ones from being sent in from remote locations
- NDP defines 5 ICMPv6 messages

Router solicitation type 133	Router advertisement type 134	Neighbour solicitation type 135	Neighbour advertisement type 136	Redirect type 137

IPv6 enhancements over IPv4



1—ICMP Type = 133 (RS)

Src = link-local address (FE80::/10)

Dst = all-routers multicast address (FF02::2)

Query = please send RA

2—ICMP Type = 134 (RA)

Src = link-local address (FE80::/10)

Dst = all-nodes multicast address (FF02::1)

Data = options, subnet prefix, lifetime, autoconfig flag

Can be sent periodically or as a response to RS

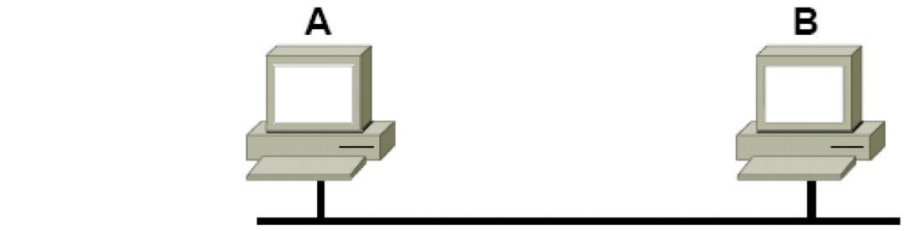
How long the host can use the router as a default router

Tells the host how addresses are assigned in the segment (via DHCP or automatic configuration)

- Router Solicitation (RS) is sent by newly-booted host to request:
  - Network prefix(es) on the segment
  - Addresses of all available default routers
  - Network Maximum Transmission Unit (MTU)
- This information is provided through the Router Advertisement (RA)

IPv6 enhancements over IPv4

- Neighbour Solicitation and Neighbour Advertisement are used to get the MAC address of a device
- It replaces ARP used by IPv4

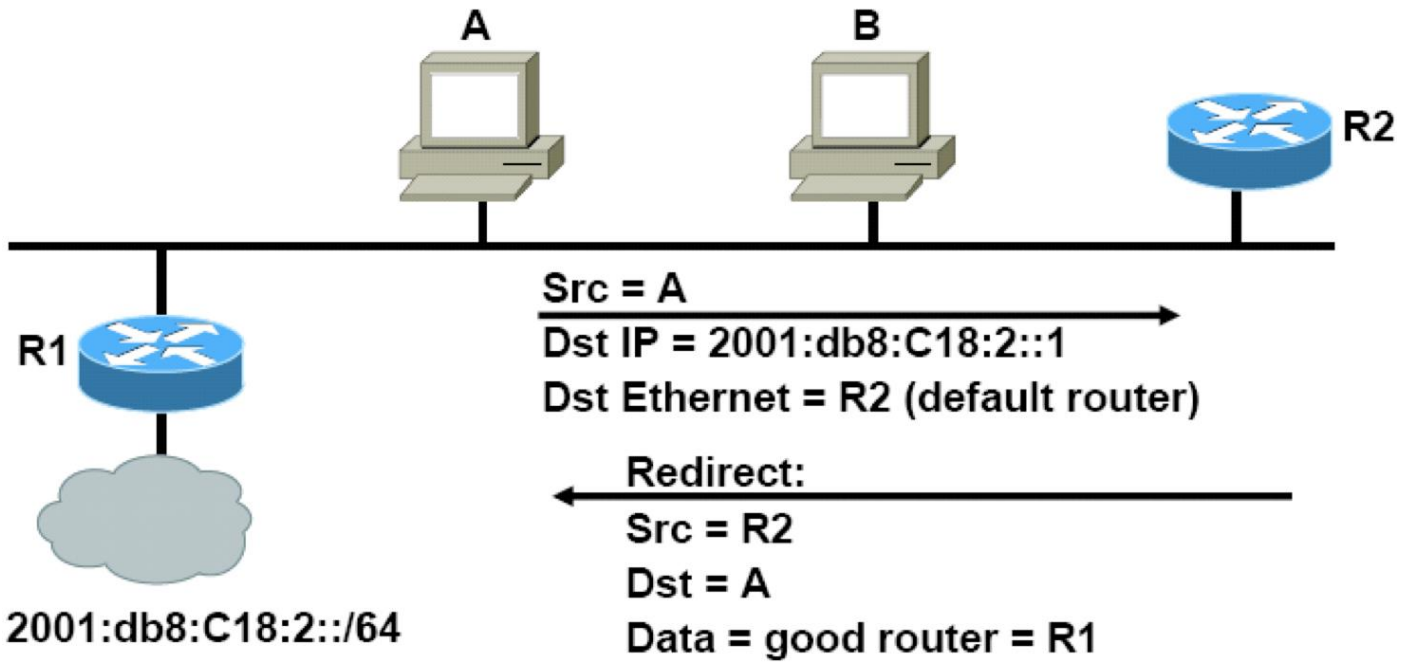


Neighbor Solicitation:
ICMP type = 135
Src = A
Dst = Solicited-node multicast address of B
Data = link-layer address of A
Query = what is your link-layer address?

Neighbor Advertisement:
ICMP type = 136
Src = B
Dst = A
Data = link-layer address of B

A and B Can Now Exchange Packets on This Link

IPv6 enhancements over IPv4



- Redirect is used by routers to signal the reroute of packets to a better router
- Hence, it's unimportant which router (of those discovered during RS phase) a host uses

THANK YOU